

In the **MaxCoverage** instances, GSEMO demonstrated fast convergence and efficient use of the available budget. The algorithm quickly identified subsets that maximised coverage in the early stages, which is expected given that the most influential elements tend to yield the largest gains at the start. After this rapid rise, the improvement curve began to flatten as the marginal benefit of adding new elements decreased, illustrating the diminishing returns property of submodular functions. The four MaxCoverage instances all followed a similar trajectory, with only small differences in their final fitness values. This consistency shows that GSEMO is resilient across different problem configurations and can handle variations in graph structure or overlap without large performance drops. By the end of the 10,000 evaluations, the algorithm had clearly stabilised, suggesting that it had effectively reached near-optimal solutions for all instances.

The **MaxInfluence** results shared the same overall trend but showed slightly higher variability during the middle stages of the runs. This was expected, as influence maximisation is inherently more complex due to the stochastic nature of influence spread across the network. During the initial evaluations, GSEMO still achieved rapid improvements by identifying highly connected hub nodes, which significantly boosted influence. However, as the search progressed, different runs explored slightly different combinations of nodes, leading to a wider spread in fitness before eventually converging. By the end of the runs, the results across all four MaxInfluence instances had stabilised to a similar range, showing that GSEMO can reliably manage the uncertainty in influence propagation while maintaining robust overall performance.

The **PackWhileTravel** instances presented a more challenging landscape. Here, GSEMO's progress was steadier and less steep, with wider variance across runs. This is likely due to the inherently multi-faceted nature of the problem, where the algorithm must balance two conflicting objectives, maximising packing value and minimising travel cost, within a more constrained and rugged search space. GSEMO's behaviour in these instances indicates that it spent more evaluations preserving a diverse set of trade-off solutions before convergence, reflecting the algorithm's ability to maintain population diversity and prevent premature stagnation. Although the overall rate of improvement was slower, the final results still showed meaningful gains and stable convergence patterns. This demonstrates that GSEMO can adapt to more complex, constraint-driven problem settings, though at the cost of a slower convergence rate.